

WASP-140b

R-2021

Benchmark run · workflow W8-benchmark-deep · record deep-bench_WASP-140_260115 · created 2026-08-08T00:06:59+00:00

BENCHMARK: INACCURATE

Results

Mid-Transit Time (Tmid)	2461055.6236 +/- 0.0037 BJD_TDB
Ratio of Planet to Stellar Radius (Rp/R*)	0.1292 +/- 0.0079
Transit depth (Rp/Rs)^2	1.67 +/- 0.2 [%]
Orbital Inclination (inc)	87.81 +/- 0.63
Airmass coefficient 1 (a1)	1.0495 +/- 0.0081
Airmass coefficient 2 (a2)	-0.029 +/- 0.024
Scatter in the residuals of the lightcurve fit is	0.77 %
Best Comparison Star	None
Optimal Aperture	4.26
Optimal Annulus	10.29
Transit Duration (day)	0.0879 +/- 0.0038

Accuracy vs published ephemeris (ExoClock/NEA)

Rp/R $\blacksquare$ fit vs published	0.1292 $\pm$ 0.0079 vs 0.1432 (deviation 1.76 σ)
Duration fit vs published	0.0879 d vs 0.0604 d (deviation 7.18 σ)
Mid-time O-C	-10.7 min
Depth SNR	16.2
Residual scatter	0.77 %

Light curve

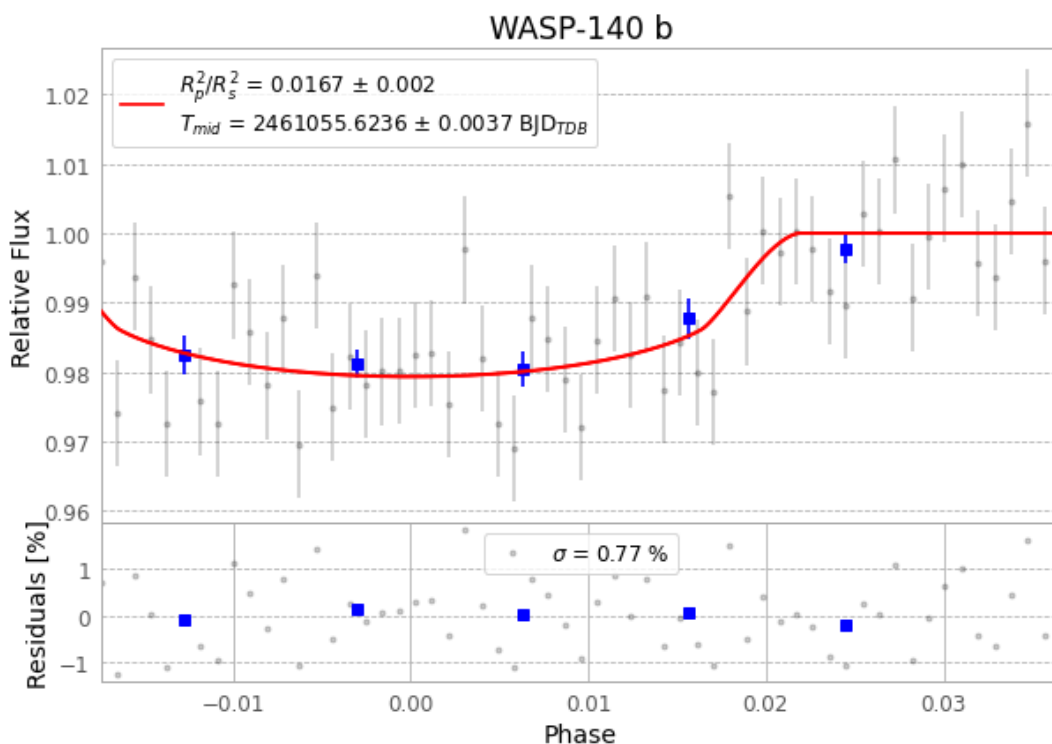


Figure 1 — FinalLightCurve_WASP-140 b_2026-01-14.png (SHA-256 14c6b0b01bc8abba...)

Method & software

EXOTIC 4.3.1 aperture photometry with nested-sampling transit fit (ultranest 3.6.5); priors from the NASA Exoplanet Archive. Camera CCD, binning 1x1, filter CV, target pixel [366, 286], 3 comparison star(s). Exact configuration: parameters.inits_json in record.json (SHA-256 113485cb0607870f...).

exotic 4.3.1 · astropy 6.1.7 · photutils 2.0.2 · numpy 1.26.4 · scipy 1.14.1 · twirl 0.5.1 · astroquery 0.4.11 · ultranest 3.6.5 · python 3.12.13 · pipeline_commit fd74a91

Data provenance

59 FITS frames (39 MB), WASP-140260115015715.FITS → WASP-140260115045116.FITS. Every input frame is SHA-256-fingerprinted in record.json; raw frames available on request and verifiable against that manifest. Artifacts: bench-WASP-140-260115.log (149e97c5c6d0...), FinalLightCurve_WASP-140 b_2026-01-14.png (14c6b0b01bc8...), FinalLightCurve_WASP-140 b_2026-01-14.csv (1dc15c8a9bfa...)

Compute resources

Wall time 251.5 s · peak memory None MB · host mac-mini-m1-8gb (macOS-26.4-arm64-arm-64bit)

Observer: — · AAVSO obscode ABGA · Automated pipeline with human review — nothing is submitted without a human approving this specific result. Data license CC-BY-4.0. Generated 2026-08-08 00:07Z.